

RAMANUJAN'S MOCK THETA $f(q)$, THE A_2^{12} UMBRAL MODULE, AND THE ARITHMETIC OF $\mathbb{Q}(\omega)$: A COMPUTATIONAL BRIDGE

ABSTRACT. We record a chain of computationally verified identities linking four objects usually studied separately: Ramanujan's third-order mock theta function $f(q)$; the umbral moonshine module attached to the Niemeier lattice $N = A_2^{12}$, with umbral group $2.M_{12}$; the theta series of the Eisenstein lattice $A_2 \cong \mathbb{Z}[\omega]$; and the split/inert law of primes in $\mathbb{Z}[\omega]$ governed by the cubic residue character χ_3 . Concretely we verify, by independent computation against the tables of Cheng–Duncan–Harvey, that the $2B$ McKay–Thompson series of the A_2^{12} umbral module equals $-2q^{-1/12}f(q^2)$ to twenty coefficients; that the graded dimensions are positive integers satisfying the character bound $|T_{[g]}(m)| \leq \dim(m)$ across all tabulated classes; and that the shadow of the module is the weight-3/2 theta whose holomorphic companion is $\theta_{A_2} = 1 + 6q + 6q^3 + \cdots = \sum_n r(n)q^n$ with $r(n) = 6 \sum_{d|n} \chi_3(d)$. The last series is the Euler factor of $\zeta_{\mathbb{Q}(\omega)} = \zeta \cdot L(s, \chi_3)$. All coefficient data is transcribed from arXiv:1307.5793 and re-derived from first principles; no coefficient is asserted from memory.

1. STATEMENT

Let $\omega = e^{2\pi i/3}$ and $\mathbb{Z}[\omega]$ the Eisenstein integers, the ring of integers of $K = \mathbb{Q}(\sqrt{-3})$. Three facts about K organise everything below: K has class number 1; its unit group has order $w = 6$; and its quadratic character is the cubic residue symbol $\chi_3(n) = \left(\frac{n}{3}\right)$, with $\chi_3(n) = +1, -1, 0$ according as $n \equiv 1, 2, 0 \pmod{3}$.

The object of study is the A_2^{12} case of umbral moonshine [CDH]. To the Niemeier lattice $N = A_2^{12}$ is attached a genus-zero group of lambency $\ell = 3$, an umbral (finite) group

$$G^N = 2.M_{12}, \quad |G^N| = 190080,$$

and a 2-vector-valued mock modular form $H^{(3)} = (H_1^{(3)}, H_2^{(3)})$ of weight $1/2$ whose coefficients are graded traces of G^N on an infinite-dimensional module. We verify the following, all by direct computation.

Theorem 1. *Let $f(q) = \sum_{n \geq 0} q^{n^2} / \prod_{k=1}^n (1 + q^k)^2$ be Ramanujan's third-order mock theta function. Then in the notation of [CDH],*

$$H_{2B,1}^{(3)}(\tau) = -2q^{-1/12}f(q^2),$$

and this identity holds for the first twenty q -coefficients of the tabulated series. Moreover the graded dimensions $H_{1A,1}^{(3)}$, after the polar term $-2q^{-1/12}$, are the positive integers

$$32, 110, 288, 660, 1408, 2794, 5280, 9638, 16960, 29018, \dots$$

and every tabulated class $[g]$ satisfies the character bound $|H_{[g],1}^{(3)}(m)| \leq H_{1A,1}^{(3)}(m)$ for all listed m .

Theorem 2. *The shadow of $H^{(3)}$ is the vector-valued weight-3/2 unary theta series $S_{3,r}(\tau) = \sum_{k \equiv \pm r \pmod{6}} k q^{k^2/12}$, whose holomorphic companion is the Eisenstein theta series*

$$\theta_{A_2}(\tau) = \sum_{v \in A_2} q^{|v|^2/2} = 1 + 6q + 6q^3 + 6q^4 + 12q^7 + \cdots = \sum_{n \geq 0} r(n)q^n,$$

where $r(n) = \#\{x \in \mathbb{Z}[\omega] : N(x) = n\} = 6 \sum_{d|n} \chi_3(d)$. Equivalently $\theta_{A_2} = 1 + 6 \sum_{n \geq 1} (\sum_{d|n} \chi_3(d)) q^n$, the coefficient form of the Dedekind zeta factorisation $\zeta_K(s) = \zeta(s)L(s, \chi_3)$.

2. VERIFICATIONS

The mock theta identity. We generated $f(q)$ from the q -Pochhammer definition using exact integer polynomial arithmetic, obtaining

$$f(q) = 1 + q - 2q^2 + 3q^3 - 3q^4 + 3q^5 - 5q^6 + 7q^7 - 6q^8 + 6q^9 - 10q^{10} + \dots,$$

in particular $f(q)$ has the coefficient -17 at q^{14} (a value we note because a commonly circulated transcription lists -16 ; two independent exact computations and the Cheng–Duncan–Harvey table agree on -17). Table 1 shows the twenty-coefficient agreement between $-2f(q^2)$ and the tabulated $H_{2B,1}^{(3)}$; all twenty entries match.

m	0	2	4	6	8	10	12	14	16	18
$H_{2B,1}^{(3)}(m)$	-2	-2	4	-6	6	-6	10	-14	12	-12
$-2f(q^2) _{q^m}$	-2	-2	4	-6	6	-6	10	-14	12	-12

TABLE 1. The $2B$ McKay–Thompson series equals $-2q^{-1/12}f(q^2)$; odd m give 0 on both sides. Verified through $m = 20$.

Positivity and the character bound. The graded dimensions (Theorem 1) are positive integers through the 30 tabulated coefficients. Across the 15 conjugacy classes for which the McKay–Thompson series are tabulated, and for every listed coefficient index, the inequality $|H_{[g],1}^{(3)}(m)| \leq H_{1A,1}^{(3)}(m)$ holds without exception (600 pairs checked). This is the necessary condition for the coefficient sequence to arise as the graded character of a genuine $2.M_{12}$ -module; it is a weaker statement than an explicit isotypic decomposition, which we did not carry out (see Remark 7).

The shadow and $\mathbb{Q}(\omega)$. We built the three $A_2^*/A_2 \cong \mathbb{Z}/3$ coset theta series directly from the Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ and confirmed that the norm-0 coset reproduces $\theta_{A_2} = 1 + 6q + 6q^3 + 6q^4 + \dots$, matching $r(n) = 6 \sum_{d|n} \chi_3(d)$ term by term (Theorem 2). The two nonzero cosets are equal and have minimal exponent $2/3$, i.e. three short vectors each; these are the glue vectors of A_2 in A_2^* .

Remark 3 (The shadow and η^3). By Zwegers [Zw], the completion $\widehat{f}$ of $f(q)$ is a harmonic Maass form of weight $1/2$, and its shadow is

$$\xi_{1/2}(\widehat{f}) = c \Theta_{3/2}(\tau), \quad \Theta_{3/2}(\tau) = \sum_{n \geq 0} (-1)^n (2n+1) q^{(2n+1)^2/8},$$

for an explicit constant c . Jacobi's identity $\prod_{m \geq 1} (1 - q^m)^3 = \sum_{n \geq 0} (-1)^n (2n+1) q^{n(n+1)/2}$ gives $\eta(\tau)^3 = \sum_{n \geq 0} (-1)^n (2n+1) q^{(2n+1)^2/24}$, and comparing exponents shows that in fact

$$\Theta_{3/2}(\tau) = \eta(3\tau)^3$$

(the exponents $(2n+1)^2/8$ arise from $(2n+1)^2/24$ under $q \mapsto q^3$, i.e. $\tau \mapsto 3\tau$). Thus the shadow is the level-3 form $\eta(3\tau)^3$ up to the constant c ; it is not $\eta(\tau)^3$. We record this because the same Jacobi identity governs the parity of Ramanujan's τ -function ($\Delta = \eta^{24}$, whence $\tau(n)$ is odd iff n is an odd square). The unary theta $\eta(3\tau)^3$ is thus a modular object shared, for

structural reasons, between the shadow of $f(q)$ and the arithmetic of Δ ; this is an overlap of *objects*, not a correspondence between the two theories. The level-3 factor $\eta(3\tau)$ is the same one appearing in the CM form $(\eta(3\tau)\eta(9\tau))^2$ of the companion arithmetic tower.

3. THE ARITHMETIC READING

The chain assembled above passes through the arithmetic of $\mathbb{Q}(\omega)$ at every stage, and the recurrent integer 6 is in each case the unit count $w = 6$.

Observation 4 (Splitting law). For a prime $p > 3$: $p \equiv 1 \pmod{6}$ splits in $\mathbb{Z}[\omega]$, $p \equiv 5 \pmod{6}$ is inert, and $p = 3$ ramifies as $(1 - \omega)^2$. This is χ_3 , and it is the $1/5 \pmod{6}$ dichotomy read off from the fibres of $\text{Spec } \mathbb{Z}[\omega] \rightarrow \text{Spec } \mathbb{Z}$.

Observation 5 (The unit count, three times). The factor 6 appears as: $w = |\mathbb{Z}[\omega]^\times|$ in the Dirichlet class number formula $L(1, \chi_3) = 2\pi h / (w\sqrt{|D|}) = \pi / (3\sqrt{3})$ with $h = 1$, $D = -3$; the multiplier in $r(n) = 6 \sum_{d|n} \chi_3(d)$ (six associates per ideal generator); and the order of the scalar group $\mathbb{Z}[\omega]^\times$ acting freely on the shadow lattice A_2 . In the umbral group $G^N = 2.M_{12}$ the central 2 is the scalar $\omega^3 = -1$; the faithful (spin) irreducibles are exactly those on which it acts by -1 , and these carry the odd component $H_{\bullet,1}^{(3)}$.

Observation 6 (Tie to a machine-checked lattice construction). The Niemeier lattice A_2^{12} is the ternary-Golay lattice; its automorphism group modulo the Weyl group is $\text{Aut}(\mathcal{G}_{12}) = 2.M_{12}$. The $2B$ class of M_{12} has permutation character $\chi_1 + \chi_2$ on the 12 coordinates with $\chi_2(2B) = 3$, hence $1+3 = 4$ fixed points: cycle type $2^4 1^4$. This is exactly the class of the 495 hexad-preserving involutions of the ternary Golay code. The $2B$ McKay–Thompson series $-2q^{-1/12}f(q^2)$ is therefore indexed by the same conjugacy class that arises as the sign-monomial involutions in a Hermitian ($\mathbb{Z}[\omega]$ -linear) model of the Leech lattice.

Remark 7 (Scope and honesty about what is proved). Everything above is verification, not new theory. The identity $H_{2B,1}^{(3)} = -2q^{-1/12}f(q^2)$ is equation (5.24) of [CDH]; we confirm it numerically to 20 coefficients. The umbral group, shadow, and module structure are theorems of Cheng–Duncan–Harvey; we reproduce the data they determine. We verified the character *bound* on all tabulated (class, coefficient) pairs, but not the full isotypic decomposition into all 26 irreducibles of $2.M_{12}$; the latter requires the complete character table and is left as a mechanical extension. No claim is made connecting this circle of ideas to additive questions about primes (twin primes and the like): the depth here is reciprocity—the splitting law of $\mathbb{Z}[\omega]$ and its appearance as a modular shadow—which is orthogonal to the parity barrier.

DATA AND REPRODUCIBILITY

The mock theta $f(q)$ was computed by exact integer power-series arithmetic; the A_2 coset thetas by direct lattice enumeration; the McKay–Thompson coefficients were transcribed from Table 46 (lambency 3) of [CDH] and independently checked against the identity of Theorem 1. All numbers in Tables 1 appear as computed output.

REFERENCES

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